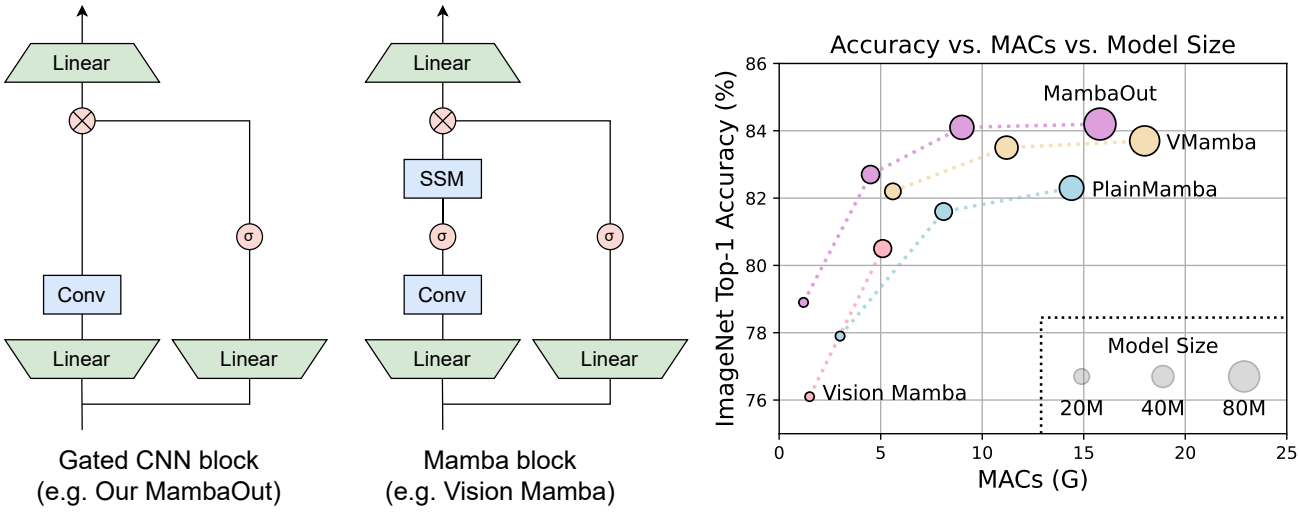
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MambaOut: Do We Really Need Mamba for Vision?*

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Code: https://github.com/yuweihao/MambaOut



(a) (b)

Figure 1. (a) Architecture of Gated CNN [20] and Mamba [28] blocks (omitting Normalization and shortcut). The Mamba block extends the Gated CNN with an additional state space model (SSM). As will be conceptually discussed in Section 3, SSM is not necessary for image classification on ImageNet [21, 72]. To empirically verify this claim, we stack Gated CNN blocks to build a series of models named *MambaOut*. (b) MambaOut outperforms visual Mamba models, *e.g.*, Vision Mamhba [112], VMamba [56] and PlainMamba [96], on ImageNet image classification.

Abstract

Mamba, an architecture with RNN-like token mixer of state space model (SSM), was recently introduced to address the quadratic complexity of the attention mechanism and subsequently applied to vision tasks. Nevertheless, the performance of Mamba for vision is often underwhelming when compared with convolutional and attention-based models. In this paper, we delve into the essence of Mamba, and conceptually conclude that Mamba is ideally suited for tasks with long-sequence *and* autoregressive *characteristics. For vision tasks, as image classification on ImageNet does not align with either characteristic, we hypothesize that Mamba is not necessary for this task; Detection and segmentation tasks on COCO or ADE20K are also not* autoregressive*, yet they adhere to the* long-sequence *characteristic, so we believe it is still worthwhile to explore Mamba's potential for these tasks. To empirically verify our hypotheses, we construct a series of models named* MambaOut *through stacking Mamba blocks while removing their core token mixer, SSM. Experimental results strongly support our hypotheses. Specifically, our MambaOut model surpasses all visual Mamba models on ImageNet image classification, indicating that Mamba is indeed unnecessary for this task. As for detection and segmentation, MambaOut cannot match the performance of state-of-the-art visual Mamba models, demonstrating the potential of Mamba for long-sequence visual tasks.*

*"MambaOut" was given to denote the model, i.e., Mamba with the SSM component taken out, rather than to imply a definitive conclusion. The vision tasks we discuss in this paper are only limited to common ones including image classification on ImageNet [21, 72], object detection & instance segmentation on COCO [53] and semantic segmentation on ADE20K [111].

1. Introduction

In recent years, Transformer [82] has become the mainstream backbone for various tasks, underpinning numerous prominent models such as BERT [22], GPT series [1, 8, 66, 67] and ViT [26]. However, the token mixer of Transformer, attention [3], incurs a quadratic complexity with respect to sequence length, posing major challenges for long sequences. To address this issue, a variety of token mixers with linear complexity to token length have been introduced [78], such as dynamic convolution [44, 90, 92], Linformer [84], Longformer [7], Big Bird [105], and Performer [14]. More recently, a new wave of RNN-like models has emerged [28, 29, 45, 65, 106], drawing significant interest from the community for their capability of parallelizable training and performing efficient inference on long sequences. Notably, models like RWKV [65] and Mamba [28] are proven to be effective as the backbone for large language models (LLMs) [52, 65].

Motivated by the promising capabilities of RNN-like models, various research endeavors have attempted to introduce Mamba [28] into visual recognition tasks [5], exemplified by the pioneering works of Vision Mamba [112], VMamba [56], LocalMamba [41], and PlainMamba [96], *etc*. The token mixer of Mamba is the structured state space models (SSM) [28–30], under the spirit of RNN. Nevertheless, their experiments show that the SSM based models for vision, in reality, lead to underwhelming performance compared with state-of-the-art convolutional [23, 31, 39, 54, 58, 70, 86, 98, 100, 101] and attention-based models [19, 24, 51, 76, 80, 81, 100, 103]. This gives rise to a compelling research question: *Do we really need Mamba for Vision?*

In this paper, we investigate the nature of Mamba, and conceptually summarize that Mamba is ideally suited for tasks with two key characteristics: *long-sequence* and *autoregressive*, because of the inherent RNN mechanism of SSM [28–30] (see explanation of Figure 2 and Figure 3). Unfortunately, not many vision tasks possess both characteristics. Image classification on ImageNet, for example, conforms to neither, while object detection & instance segmentation on COCO and semantic segmentation on ADE20K conform only to the *long-sequence*. *Autoregressive* characteristic, on the other hand, demands that each token aggregate information solely from preceding and current tokens, a concept denoted as *causal mode* for token mixing [69] (see Figure 3(a)). In fact, all visual recognition tasks fall within the understanding domain rather than the generative one, meaning that the model can see the entire image at once. As such, imposing additional causal constraints on token mixing in visual recognition models could lead to a performance drop (see Figure 3(b)). Although this issue can be mitigated via bidirectional branches [74], it is inevitable that the issue persists within each branch.

Based on the conceptual discussion above, we propose the two hypotheses as follows:

- *Hypothesis 1*: SSM is not necessary for ImageNet image classification, since this task conforms to neither the *long-sequence* or *autoregressive* characteristic.
- *Hypothesis 2*: SSM may be potentially beneficial for object detection & instance segmentation on COCO and semantic segmentation on ADE20K, since they follow the *long-sequence* characteristic, though they are not *autoregressive*.

To experimentally validate our hypotheses, we developed a series of models termed *MambaOut* through stacking Gated CNN [20] blocks. The key distinction between Gated CNN and Mamba blocks lies in the existence of SSM, as illustrated in Figure 1(a). Experimental results demonstrate that the simpler MambaOut model, in reality, already surpasses the performance of visual Mamba models [41, 56, 96, 112], which in turn verifies our *Hypothesis 1*. We also show empirical results that MambaOut falls short of matching the performance of state-of-the-art visual Mamba models [41, 56] in detection and segmentation tasks (see Tables 2 and 3), which underscores the potential of SSM on these tasks and effectively validates our *Hypothesis 2*.

The contributions of our paper are threefold. Firstly, we analyze the RNN-like mechanism of SSM and conceptually conclude that Mamba is suited for tasks with long-sequence and autoregressive characteristics. Secondly, we examine the characteristics of visual tasks and hypothesize that SSM is unnecessary for image classification on ImageNet since this task does not meet either characteristic, yet exploring the potential of SSM for detection and segmentation tasks remains valuable since these tasks conform to long-sequence characteristic, though they are not autoregressive. Thirdly, we develop a series of models named MambaOut based on Gated CNN blocks but without SSM. Experiments show that MambaOut effectively surpasses visual Mamba models in ImageNet image classification but does not reach the performance of state-of-the-art visual Mamba models in detection and segmentation tasks. These observations, in turn, validate our hypotheses. As such, MambaOut, because of its *Occam's razor* nature, may readily serve as a natural baseline for future research on visual Mamba models.

2. Related work

Transformer has been widely utilized across various domains, including BERT [22] and GPT series [1, 8, 66, 67] in NLP and ViT [26] in computer vision. However, the attention module in Transformers scales quadratically with sequence length, presenting a significant computational challenge. Numerous studies [78] have explored various strategies to mitigate this issue, including low-rank approaches [84], kernelization [14, 45], token mixing range limitation [7, 33, 57, 105], and history memory compression [68].

More recently, RNN-like methods [18, 45, 106], particularly RWKV [65] and Mamba [28], have garnered attention for their promising results in large language models [52, 65].

Eager exploratory researchers have quickly moved to incorporate SSM and Mamba [28] into visual recognition tasks [41, 49, 56, 63, 64, 94, 96, 108, 112]. For instance, Vision Mamba [112] integrates Mamba [28] to develop isotropic vision models akin to ViT [26], demonstrating its long-range modeling capability; VMamba [56] employs Mamba to construct hierarchical vision models similar to AlexNet [47] and ResNet [34]; LocalMamba [41] enhances visual Mamba models [56, 112] by incorporating local inductive biases; PlainMamba [96] aims to further enhance the performance of isotropic Mamba models; EfficientVMamba [64] focuses on efficiency through the introduction of atrous selective scan for lightweight visual Mamba models. MambaMixer [6] also utilizes SSM for channel mixing besides token mixing. One concurrent work, MILA [32], offers an alternative design by demystifying Mamba from a linear attention perspective.

Unlike these initiatives, our work does not aim to design new visual Mamba models. Instead, we explore a pertinent research question about the necessity of Mamba [28] in visual recognition contexts [21, 53, 72, 111]. We hope this paper can provide insights for future research on visual Mamba models.

3. Conceptual discussion

In this section, we first discuss what characteristics of tasks the Mamba model is suited for. Next, we examine whether visual recognition tasks conform to these characteristics. Based on the examination results, we propose hypotheses regarding the necessity of Mamba for vision.

3.1. What tasks is Mamba suitable for?

The token mixer of Mamba is selective SSM [28, 29] which defines four parameters $(\Delta, \mathbf{A}, \mathbf{B}, \mathbf{C})$ and transforms them to $(\overline{\mathbf{A}}, \overline{\mathbf{B}}, \mathbf{C})$ by

$$\overline{\mathbf{A}} = \exp(\Delta A), \quad \overline{\mathbf{B}} = (\Delta\mathbf{A})^{-1}(\exp(\Delta\mathbf{A})-\mathbf{I})\cdot\Delta\mathbf{B}. \quad (1)$$

Then the sequence-to-sequence transformation of SSM can be expressed by

$$h_t = \overline{\mathbf{A}}h_{t-1} + \overline{\mathbf{B}}x_t, \quad (2)$$
$$y_t = \mathbf{C}h_t, \quad (3)$$

where t denotes the timestep, x_t represents the input, h_t signifies the hidden state, and y_t indicates the output. The recurrent property [38] of Equation 2 distinguishes RNN-like SSM from causal attention. The hidden state h can be seen as a fixed-size memory that stores all historical information. Through Equation 2, this memory is updated while its size remains constant. The fixed size means the memory is inevitably lossy, but it ensures that the computational complexity of integrating the memory with the current input remains constant. Conversely, causal attention stores all keys and values from previous tokens as its memory, which expands by adding the current token's key and value with each new input. This memory is theoretically lossless. However, as more tokens are inputted, the memory size grows, thereby increasing the complexity of integrating the memory with the current input. The differences in memory mechanisms between RNN-like models and causal attention are further illustrated in Figure 2.

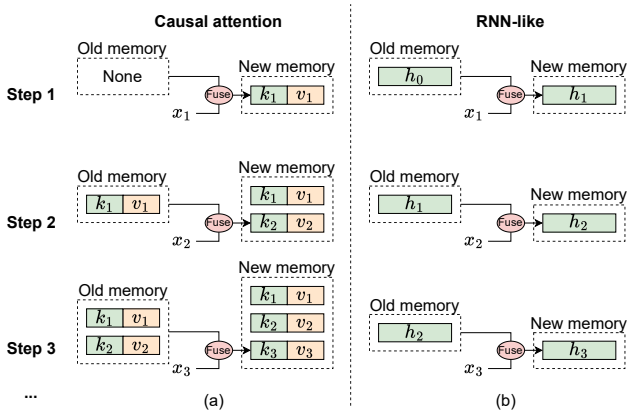
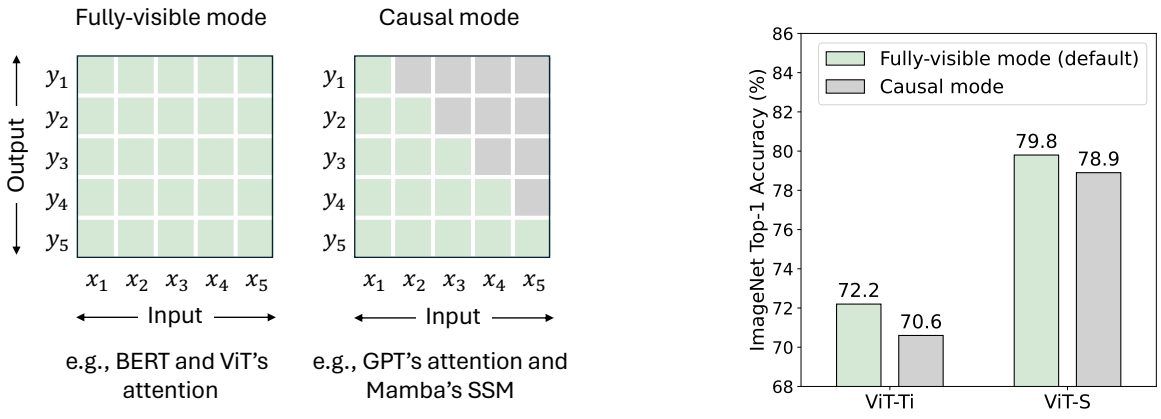


Figure 2. The mechanism illustration of causal attention and RNN-like models from memory perspective, where x_i denotes the input token of i-th step. (a) Causal attention stores all previous tokens' keys k and values v as memory. The memory is updated by continuously adding the current token's key and value, so the memory is lossless, but the downside is that the computational complexity of integrating old memory and current tokens increases as the sequence lengthens. Therefore, attention can effectively manage short sequences but may encounter difficulties with longer ones. (b) In contrast, RNN-like models compress previous tokens into fixed-size hidden state h, which serves as the memory. This fixed size means that RNN memory is inherently lossy, which cannot directly compete with the lossless memory capacity of attention models. Nonetheless, **RNN-like models can demonstrate distinct advantages in processing long sequences, as the complexity of merging old memory with current input remains constant, regardless of sequence length.**

Because SSM's memory is inherently lossy, it logically falls short of the lossless memory of attention. Consequently, Mamba cannot showcase its strengths in handling short sequences, an area where attention performs well with ease. However, in scenarios involving long sequences, attention will falter due to its quadratic complexity. In this case, Mamba can distinctly highlight its efficiency in merging memory with the current input, thus managing long sequences smoothly. Therefore, Mamba is particularly well-suited for processing long sequences.

Although the recurrent nature of SSM (Equation 2) allows Mamba to handle long sequences efficiently, it intro-



(a) (b)

Figure 3. (a) Two modes of token mixing [69]. For a total of T tokens, the fully-visible mode allows token t to aggregate inputs from all tokens, i.e., $\{xi\}_{i=1}^{T}$, to compute its output y_t. In contrast, the causal mode restricts token t to only aggregate inputs from preceding and current tokens $\{x_i\}_{i=1}^{t}$. By default, attention operates in fully-visible mode but can be adjusted to causal mode with causal attention masks. RNN-like models, such as Mamba's SSM [28, 29], inherently operate in causal mode due to their recurrent nature. (b) **We modify the ViT's attention [26, 79] from fully-visible to causal mode and observe performance drop on ImageNet, which indicates causal mixing is unnecessary for understanding tasks.**

duces a significant limitation: h_t can only access information from the previous and current timesteps. As illustrated in Figure 3, this type of token mixing is termed causal mode, which can be formulated as:

$$y_t = f(x_1, x_2, ..., x_t), \tag{4}$$

where x_t and y_t represent the input and output of the t-th token, respectively. Due to its causal nature, this mode is well-suited for autoregressive generation tasks.

Another mode is called fully-visible mode, where each token can aggregate information from all preceding and subsequent tokens. This means the output of each token depends on the inputs from all tokens:

$$y_t = f(x_1, x_2, ..., x_t, ..., x_T), \tag{5}$$

where T represents the total number of tokens. The fully-visible mode is suitable for understanding tasks, where all inputs can be accessed by the model at once.

Attention is in fully-visible mode by default, but it can easily turn into causal mode by applying causal masks to the attention maps. RNN-like models inherently operate in causal mode due to their recurrent properties, as illustrated by Mamba's Equation 2. Due to this inherent characteristic, RNN-like models cannot be transformed into fully-visible mode. Although RNNs can approximate a fully-visible mode using bidirectional branches, each branch still individually remains in causal mode. Therefore, Mamba is well-suited for tasks that require causal token mixing, due to the inherent limitations of its recurrent properties.

In summary, Mamba is ideally suited for tasks that display the following characteristics:

- *Characteristic 1*: The task involves processing long sequences for global modeling.[1]
- *Characteristic 2*: The task requires causal token mixing mode.

Next, we will discuss whether visual recognition tasks exhibit these two characteristics.

3.2. Do visual recognition tasks have very long sequences?

In this subsection, we explore whether visual recognition tasks necessitate long sequence modeling. We use the Transformer model [82] as a case study to facilitate our analysis, because SSMs [28, 29] are proposed to address the Transformers' computational inefficiency on long sequence. Consider a Transformer block with a common MLP ratio of 4; assuming its input $X \in \mathbb{R}^{L \times D}$ has a token length of L and channel (embedding) dimensions of D, the FLOPs for the block can be calculated as:

$$\text{FLOPs} = 24D^2L + 4DL^2. \tag{6}$$

From this, we derive the ratio of the quadratic term to the linear term in L as:

$$r_L = \frac{4DL^2}{24D^2L} = \frac{L}{6D}. \tag{7}$$

If $L > 6D$, the computational load of the quadratic term in L surpasses that of the linear term. This provides a simple metric to determine if the task involves long sequences.[2]

[1] For scenarios requiring only local modeling, processing short sequences within windowed segments is sufficient.

[2] In this paper, this metric is simply defined based on FLOPs, but it can certainly be defined from other perspectives in different contexts.

Model	Token Mixing Type	Param (M)	Test@224^2 MAC (G)	Test@224^2 Acc (%)
VAN-B0 [31]	Conv	4	0.9	75.4
MogaNet-T [50]	Conv	5	1.1	**79.0**
FasterNet-T1 [9]	Conv	8	0.9	76.2
InceptionNeXt-A [101]	Conv	4	0.5	75.3
DeiT-Ti [79]	Attn	6	1.3	72.2
T2T-ViT-7 [102]	Attn	4	1.1	71.7
PVTv2-B0 [85]	Conv + Attn	3	0.6	70.5
MobileViTv3-XS [83]	Conv + Attn	3	0.9	76.7
EMO-6M [109]	Conv + Attn	6	1.0	**79.0**
Vim-Ti [112]	Conv + SSM	7	1.5	76.1
LocalVim-T [41]	Conv + SSM	8	1.5	76.2
EfficientVMamba-T [64]	Conv + SSM	6	0.8	76.5
EfficientVMamba-S [64]	Conv + SSM	11	1.3	78.7
MambaOut-Femto	Conv	7	1.2	**78.9**
PoolFormer-S24 [99]	Pool	21	3.4	80.3
ConvNeXt-T [58]	Conv	29	4.5	82.1
VAN-B2 [31]	Conv	27	5.0	82.8
ConvFormer-S18 [100]	Conv	27	3.9	83.0
MogaNet-S [50]	Conv	25	5.0	83.4
InternImage-T [86]	Conv	30	5	83.5
InceptionNeXt-T [101]	Conv	28	4.2	82.3
DeiT-S [79]	Attn	22	4.6	79.8
T2T-ViT-14 [102]	Attn	22	4.8	81.5
Swin-T [57]	Attn	29	4.5	81.3
Focal-Tiny [97]	Attn	29	4.9	82.2
CSWin-T [24]	Attn	23	4.3	82.7
CoAtNet-0 [19]	Conv + Attn	25	4.2	81.6
iFormer-S [76]	Conv + Attn	20	4.8	83.4
MOAT-0 [95]	Conv + Attn	28	5.7	83.3
CAFormer-S18 [100]	Conv + Attn	26	4.1	83.6
SG-Former-S [71]	Conv + Attn	23	4.8	83.2
TransNeXt-Tiny [75]	Conv + Attn	28	5.7	**84.0**
Vim-S [112]	Conv + SSM	26	5.1	80.5
VMamba-T [56]	Conv + SSM	22	5.6	82.2
Mamba-2D-S [49]	Conv + SSM	24	–	81.7
LocalVim-S [41]	Conv + SSM	28	4.8	81.2
LocalVMamba-T [41]	Conv + SSM	26	5.7	82.7
EfficientVMamba-B [64]	Conv + SSM	33	4.0	81.8
PlainMamba-L1 [96]	Conv + SSM	7	3.0	77.9
VMambaV3-T* [55]	Conv + SSM	31	4.9	82.6
MambaOut-Tiny	Conv	27	4.5	**82.7**

Model	Token Mixing Type	Param (M)	Test@224^2 MAC (G)	Test@224^2 Acc (%)
ConvNeXt-S [58]	Conv	50	8.7	83.1
VAN-B3 [31]	Conv	45	9.0	83.9
ConvFormer-S36 [100]	Conv	40	7.6	84.1
InternImage-S [86]	Conv	50	8	84.2
MogaNet-B [50]	Conv	44	9.9	84.3
T2T-ViT-19 [102]	Attn	39	8.5	81.9
Swin-S [57]	Attn	50	8.7	83.0
Focal-Small [97]	Attn	51	9.1	83.5
CSWin-S [24]	Attn	35	6.9	83.6
MViTv2-S [51]	Attn	35	7.0	83.6
CoAtNet-1 [19]	Conv + Attn	42	8.4	83.3
UniFormer-B [48]	Conv + Attn	50	8.3	83.9
CAFormer-S36 [100]	Conv + Attn	39	8.0	84.5
SG-Former-M [71]	Conv + Attn	39	7.5	84.1
TransNeXt-Small [75]	Conv + Attn	50	10.3	**84.7**
VMamba-S [56]	Conv + SSM	44	11.2	83.5
LocalVMamba-S [41]	Conv + SSM	50	11.4	83.7
PlainMamba-L2 [96]	Conv + SSM	25	8.1	81.6
VMambaV3-S* [55]	Conv + SSM	50	8.7	83.6
MambaOut-Small	Conv	48	9.0	**84.1**
ConvNeXt-B [58]	Conv	89	15.4	83.8
RepLKNet-31B [23]	Conv	79	15.3	83.5
ConvFormer-M36 [100]	Conv	57	12.8	84.5
HorNet-B [70]	Conv	88	15.5	84.3
MogaNet-L [50]	Conv	83	15.9	84.7
InternImage-B [86]	Conv	97	16	84.9
DeiT-B [79]	Attn	86	17.5	81.8
T2T-ViT-24 [102]	Attn	64	13.8	82.3
Swin-B [57]	Attn	88	15.4	83.5
CSwin-B [24]	Attn	78	15.0	84.2
MViTv2-B [51]	Attn	52	10.2	84.4
CoAtNet-2 [19]	Conv + Attn	75	15.7	84.1
iFormer-L [76]	Conv + Attn	87	14.0	84.8
MOAT-2 [95]	Conv + Attn	73	17.2	84.7
CAFormer-M36 [100]	Conv + Attn	56	13.2	**85.2**
TransNeXt-Base [75]	Conv + Attn	90	18.4	84.8
VMamba-B [56]	Conv + SSM	75	18.0	83.7
Mamba-2D-B [49]	Conv + SSM	92	–	83.0
PlainMamba-L3 [96]	Conv + SSM	50	14.4	82.3
VMambaV3-B* [55]	Conv + SSM	89	15.4	83.9
MambaOut-Base	Conv	85	15.8	**84.2**

Table 1. **Performance of models on ImageNet at the resolution of 224^2.** Our MambaOut model employs the Gated CNN block [20]. The Mamba block [28], derived from the Gated CNN block, incorporates an additional SSM (state space model). It is evident that visual Mamba models fall short of MambaOut's performance, let alone surpassing state-of-the-art convolutional or convolution-attention-hybrid models. * Note that VMambaV3 modifies the meta-architecture of the Mamba block to MetaFormer [100], different from other visual Mamba models and MambaOut.

For instance, with 384 channels in ViT-S, the threshold $\tau_{\text{small}} = 6 \times 384 = 2304$, and for 768 channels in ViT-B, $\tau_{\text{base}} = 6 \times 768 = 4608$.

For image classification on ImageNet, the typical input image size is 224^2, resulting in $14^2 = 196$ tokens with patch size of 16^2.[3] Clearly, 196 is much less than both τ_{small} and τ_{base}, indicating that image classification on ImageNet does not qualify as a long-sequence task.

For object detection & instance segmentation on COCO, with an inference image size of 800×1280, and for semantic segmentation on ADE20K, with an inference image size of 512×2048, the number of tokens is approximately 4K, given patch size of 16^2. Since $4K > \tau_{\text{small}}$ and

[3]For hierarchical 4-stage architecture, its main part is the third stage, where the downsampling ratio is also 16^2, the same as ViT. The first and second stages focus on local modeling, serving a role similar to ViT's patch embedding. Therefore, processing short sequences within windowed segments is sufficient at these stages.

$4K \approx \tau_{\text{base}}$, both detection on COCO and segmentation on ADE20K can be considered long-sequence tasks.

3.3. Do visual recognition tasks need causal token mixing mode?

As discussed in Section 3.1 and illustrated in Figure 3, the fully-visible token mixing mode allows unrestricted range of mixing, whereas the causal mode limits the current token to only access information from preceding tokens. Visual recognition is categorized as an understanding task, wherein the model can see the entire image at once, eliminating the need for restrictions on token mixing. Imposing additional constraints on token mixing can potentially degrade model performance. As demonstrated in Figure 3(b), when causal restrictions are applied to Vision Transformers (ViT) [26, 79], a noticeable decline in performance is observed. Generally, the fully-visible mode is appropriate for understanding tasks, while the causal mode is better suited for autoregressive tasks. This claim can also be substantiated by the observation that BERT [22] and ViT [26] (BEiT [4] and MAE [36]) are used more for understanding tasks than GPT-1/2 [66, 67] and image GPT [11]. Therefore, visual recognition tasks do not need causal token mixing mode.

3.4. Hypotheses regarding the necessity of Mamba for vision

Based on our preceding discussion, we summarize our hypotheses regarding the necessity of introducing Mamba for visual recognition tasks as follows:

- *Hypothesis 1*: It is not necessary to introduce SSM for image classification on ImageNet, as this task does not meet *Characteristic 1* or *Characteristic 2*.
- *Hypothesis 2*: It is still worthwhile to further explore the potential of SSM for visual detection and segmentation on COCO or ADE20K since these tasks align with *Characteristic 1*, despite not fulfilling *Characteristic 2*.

4. Experimental verification

4.1. Gated CNN and MambaOut

Next, we aim to validate our hypotheses empirically. As depicted in Figure 1(a), Mamba block [28] is based on the Gated CNN block [20]. The meta-architecture of Gated CNN and Mamba can be considered as a simplified integration of the MetaFormer's [99] token mixer and an MLP, akin to MetaNeXt [101]. Formally, given the input $X \in \mathbb{R}^{N \times D}$, the meta-architecture is formulated as:

$$X' = \text{Norm}(X), \tag{8}$$

$$Y = (\text{TokenMixer}(X'W_1) \odot \sigma(X'W_2))W_3 + X, \tag{9}$$

where $\text{Norm}(\cdot)$ represents normalization [2, 42, 91]; $\text{TokenMixer}(\cdot)$ refers to the module to conduct token mixing [100]; $W_1 \in \mathbb{R}^{D \times rD}$, $W_2 \in \mathbb{R}^{D \times rD}$ and $W_3 \in \mathbb{R}^{rD \times D}$ are learnable parameters with MLP expansion r; σ is activation function [27, 37]. Token mixers of Gated CNN and Mamba are:

$$\text{TokenMixer}_{\text{GatedCNN}}(Z) = \text{Conv}(Z) \tag{10}$$

$$\text{TokenMixer}_{\text{Mamba}}(Z) = \text{SSM}(\sigma(\text{Conv}(Z))) \tag{11}$$

Comparing Equations 10 and 11, and referencing Figure 1(a), the primary distinction between the Gated CNN [20] and the Mamba block [28] lies in the presence of SSM. This prompts us to develop a series of models, termed MambaOut, which are based on the Gated CNN block without SSM. MambaOut will help us assess the necessity of Mamba for visual recognition tasks.

Specifically, we specify the token mixer of Gated CNN as depthwise convolution [13] of 7×7 kernel size, following ConvNeXt [58, 61]. The ablation study of kernel size is shown in the appendix. Besides, to improve the practical speed, we only conduct depthwise convolution on partial channels [9, 60, 101], following InceptionNeXt [101]. Similar to ResNet, we adopt 4-stage framework to build MambaOut by stacking Gated CNN blocks at each stage and the configuration details of each model size are shown in the appendix. We also show isotropic MambaOut in the appendix.

4.2. Image classification on ImageNet

Setup. ImageNet [21, 72] serves as the gold standard benchmark for image classification, encompassing a wide array of 1,000 common classes. It comprises approximately 1.3 million training images and 50,000 validation images. The training scheme follows DeiT [79] without distillation. Specifically, the used data augmentation contains random resized crop (input image size of 224^2), horizontal flip, RandAugment [17], Mixup [107], CutMix [104], Random Erasing [110] and color jitter; Regularization techniques include weight decay, stochastic depth [40] and label smoothing [77]. All our models are trained by AdamW [46, 59]. The learning rate scaling rule is $\text{lr} = \frac{\text{batchsize}}{1024} \times 10^{-3}$. In this paper, we set the batch size to 4096, so the learning rate is 0.004. Our MambaOut models are implemented with PyTorch [62] and timm [88] libraries and trained on TPU v3. More training hyper-parameters are shown in the appendix.

Results. The performance of our MambaOut models, visual Mamba models, and other various convolution and attention-based models on ImageNet [21, 72] is presented in Table 1. Notably, our MambaOut models, which do not incorporate SSM, consistently outperform visual Mamba models [41, 56, 64, 96, 112] that include SSM across all model sizes. For instance, the MambaOut-Small model achieves top-1 accuracy of 84.1%, 0.4% higher than that of LocalVMamba-S [41], while requiring only 79% of the MACs. These results strongly support our *Hypothesis 1*, which posits that introducing SSM for image classification

Backbone	Token Mixing Type	Param (M)	MAC (G)	Mask R-CNN 1× schedule					
				AP^b	AP^b_{50}	AP^b_{75}	AP^m	AP^m_{50}	AP^m_{75}
ConvNeXt-T [54]	Conv	48	262	44.2	66.6	48.3	40.1	63.3	42.8
FocalNet-T [98]	Conv	49	268	46.1	68.2	50.6	41.5	65.1	44.5
Swin-T [57]	Attn	48	267	42.7	65.2	46.8	39.3	62.2	42.2
ViT-Adapter-S [12]	Attn	48	403	44.7	65.8	48.3	39.9	62.5	42.8
CSWin-T [24]	Attn	42	279	46.7	68.6	51.3	42.2	65.6	45.4
PVTv2-B2 [85]	Conv + Attn	45	309	45.3	67.1	49.6	41.2	64.2	44.4
SG-Former-S [71]	Conv + Attn	41	–	47.4	69.0	52.0	42.6	65.9	46.0
TransNeXt-Tiny [75]	Conv + Attn	48	356	**49.9**	71.5	54.9	**44.6**	68.6	48.1
VMamba-T [56]	Conv + SSM	42	286	46.5	68.5	50.7	42.1	65.5	45.3
LocalVMamba-T [41]	Conv + SSM	45	291	46.7	68.7	50.8	42.2	65.7	45.5
EfficientVMamba-B [64]	Conv + SSM	53	252	43.7	66.2	47.9	40.2	63.3	42.9
VMambaV3-T [55]	Conv + SSM	50	270	**47.4**	69.5	52.0	**42.7**	66.3	46.0
PlainMamba-L1 [96]	Conv + SSM	31	388	44.1	64.8	47.9	39.1	61.6	41.9
MambaOut-Tiny	Conv	43	262	45.1	67.3	49.6	41.0	64.1	44.1
ConvNeXt-S [54]	Conv	70	348	45.4	67.9	50.0	41.8	65.2	45.1
FocalNet-S [98]	Conv	72	365	48.3	70.5	53.1	43.1	67.4	46.2
Swin-S [57]	Attn	69	354	44.8	66.6	48.9	40.9	63.2	44.2
CSWin-S [24]	Attn	54	342	47.9	70.1	52.6	43.2	67.1	46.2
PVTv2-B3 [85]	Conv + Attn	65	397	47.0	68.1	51.7	42.5	65.7	45.7
SG-Former-M [71]	Conv + Attn	51	–	48.2	70.3	53.1	43.6	66.9	47.0
TransNeXt-Small [75]	Conv + Attn	69	516	**51.1**	72.6	56.2	**45.5**	69.8	49.1
VMamba-S [56]	Conv + SSM	64	400	48.2	69.7	52.5	43.0	66.6	46.4
LocalVMamba-S [41]	Conv + SSM	69	414	48.4	69.9	52.7	43.2	66.7	46.5
VMambaV3-S [55]	Conv + SSM	64	357	**48.7**	70.0	53.4	**43.7**	67.3	47.0
MambaOut-Small	Conv	65	354	47.4	69.1	52.4	42.7	66.1	46.2
ConvNeXt-B [54]	Conv	108	486	47.0	69.4	51.7	42.7	66.3	46.0
FocalNet-B [98]	Conv	111	507	49.0	70.9	53.9	43.5	67.9	46.7
Swin-B [57]	Attn	107	496	46.9	–	–	42.3	–	–
ViT-Adapter-B [12]	Attn	102	557	47.0	68.2	51.4	41.8	65.1	44.9
CSWin-B [24]	Attn	97	526	48.7	70.4	53.9	43.9	67.8	47.3
PVTv2-B5 [85]	Conv + Attn	102	557	47.4	68.6	51.9	42.5	65.7	46.0
TransNeXt-Base [75]	Conv + Attn	109	728	**51.7**	73.2	56.9	**45.9**	70.5	49.7
VMamba-B [56]	Conv + SSM	96	540	48.5	69.6	53.0	43.1	67.0	46.4
PlainMamba-L2 [96]	Conv + SSM	53	542	46.0	66.9	50.1	40.6	63.8	43.6
VMambaV3-B [55]	Conv + SSM	108	485	**49.2**	70.9	53.9	**43.9**	67.7	47.6
MambaOut-Base	Conv	100	495	47.4	69.3	52.2	43.0	66.4	46.3

Table 2. **Performance of object detection and instance segmentation on COCO with Mask R-CNN.** The MACs are measured with input size of 800 × 1280.

on ImageNet is unnecessary, aligning with the principle of Occam's razor.

Additionally, visual Mamba models currently exhibit a significant performance gap when compared to state-of-the-art convolution and attention models. For instance, the CAFormer-M36 [100], which employs old-fashioned token mixers of simple separable convolutions from MobileNetV2 [73] and vanilla attention from Transformer [82] invented more than 7 years ago, outperforms all visual Mamba models of comparable size by more than 1% accuracy. Should future research aim to challenge our *Hypothesis 1*, it will be necessary to develop visual Mamba models with token mixers of convolution and SSM to achieve state-of-the-art performance on ImageNet.

4.3. Object detection & instance segmentation on COCO

Setup. COCO 2017 [53] serves as a widely recognized benchmark for object detection and instance segmentation. In our experiments, MambaOut is employed as the backbone within Mask R-CNN [35], initialized with weights pre-trained on ImageNet. We adhere to the standard 1× training schedule of 12 epochs. The training images are resized such that the shorter side measures 800 pixels, while the longer side does not exceed 1333 pixels. The AdamW optimizer [46, 59] is used with a learning rate of 0.0001 and a total batch size of 16. Our implementation leverages the PyTorch [62] and mmdetection [10] libraries. We utilize FP16 precision to save training costs. The experiments are conducted on 4 GPUs of NVIDIA 4090.

Results. Although MambaOut can surpass some visual Mamba models [64, 96] in object detection and instance segmentation on COCO [53], it still lags behind the state-of-the-art visual Mambas, such as VMamba [56] and LocalVMamba [56]. For instance, the performance of MambaOut-Tiny as the backbone for Mask R-CNN trails VMamba-T [56] by 1.4 AP^b and 1.1 AP^m. This performance disparity underscores the benefits of integrating Mamba in long-sequence visual tasks, reinforcing our *Hypothesis 2*. However, visual Mamba still exhibits a significant performance gap when compared to the state-of-the-art convolution-attention-hybrid models, TransNeXt [75]. Visual Mamba needs to further validate its effectiveness by outperforming state-of-the-art models in the visual detection task.

4.4. Semantic segmentation on ADE20K

Setup. ADE20K [111], a widely-used benchmark for the semantic segmentation task, encompasses 150 semantic cat-

Backbone	Token Mixing Type	UperNet Param (M)	MAC (G)	mIoU (SS)	mIoU (MS)
ConvNeXt-T [54]	Conv	60	939	46.0	46.7
HorNet-T [70]	Conv	55	924	49.2	49.3
ConvFormer-S18 [100]	Conv	54	925	47.5	48.6
InternImage-T [86]	Conv	59	944	47.9	48.1
Swin-T [57]	Attn	60	945	44.4	45.8
Twins-S [15]	Attn	54	901	46.2	47.1
Focal-T [97]	Attn	62	998	45.8	47.0
CSWin-T [24]	Attn	60	959	49.3	50.7
UniFormer-S [48]	Conv + Attn	52	955	47.0	48.5
CAFormer-S18 [100]	Conv + Attn	54	1024	48.1	48.9
SG-Former-S [71]	Conv + Attn	53	989	49.9	51.5
TransNeXt-Tiny [75]	Conv + Attn	59	978	**51.1**	**51.2**
VMamba-T [56]	Conv + SSM	55	964	47.3	48.3
LocalVMamba-T [41]	Conv + SSM	57	970	**47.9**	**49.1**
EfficientVMamba-B [64]	Conv + SSM	65	930	46.5	47.3
PlainMamba-L2 [96]	Conv + SSM	55	285	–	46.8
PlainMamba-L3 [96]	Conv + SSM	81	419	–	**49.1**
VMambaV3-T [56]	Conv + SSM	62	949	**47.9**	48.8
MambaOut-Tiny	Conv	54	938	47.4	48.6
ConvNeXt-S [54]	Conv	82	1027	48.7	49.6
HorNet-S [70]	Conv	85	1027	50.0	50.5
ConvFormer-S36 [100]	Conv	67	1003	49.6	50.7
InternImage-S [86]	Conv	80	1017	50.1	50.9
Swin-S [57]	Attn	81	1038	47.6	49.5
Twins-B [15]	Attn	89	1020	47.7	48.9
to be continued to the right					

Backbone	Token Mixing Type	UperNet Param (M)	MAC (G)	mIoU (SS)	mIoU (MS)
continued from the left					
Focal-S [97]	Attn	85	1130	48.0	50.0
CSWin-S [24]	Attn	65	1027	50.4	51.5
CAFormer-S36 [100]	Conv + Attn	67	1197	50.6	50.8
SG-Former-M [71]	Conv + Attn	68	1114	51.2	52.1
TransNeXt-Small [75]	Conv + Attn	80	1089	**52.2**	**52.3**
VMamba-S [56]	Conv + SSM	76	1081	49.5	50.5
LocalVMamba-S [41]	Conv + SSM	81	1095	50.0	51.0
VMambaV3-S [55]	Conv + SSM	82	1028	**50.6**	**51.2**
MambaOut-Small	Conv	76	1032	49.5	50.6
ConvNeXt-B [54]	Conv	122	1170	49.1	49.9
FocalNet-B [98]	Conv	126	1192	50.5	51.4
HorNet-B [70]	Conv	126	1171	50.5	50.9
ConvFormer-M36 [100]	Conv	85	1113	50.4	51.3
InternImage-B [86]	Conv	128	1185	50.8	51.3
Swin-B [57]	Attn	121	1188	48.1	49.7
Twins-L [15]	Attn	133	1164	48.8	50.2
Focal-B [97]	Attn	126	1354	49.0	50.5
CSWin-B [24]	Attn	110	1222	51.1	52.2
UniFormer-B [48]	Conv + Attn	80	1106	49.5	50.7
CAFormer-M36 [100]	Conv + Attn	84	1346	51.7	51.7
SG-Former-B [71]	Conv + Attn	109	1304	52.0	52.7
TransNeXt-Base [75]	Conv + Attn	121	1268	**53.0**	**53.4**
VMamba-B [56]	Conv + SSM	110	1226	50.0	51.3
VMambaV3-B [55]	Conv + SSM	122	1170	**51.0**	**51.6**
MambaOut-Base	Conv	112	1178	49.6	51.0

Table 3. **Performance of Semantic segmentation with UperNet [93] on ADE20K [111] validation set.** The MACs are measured with input size of 512×2048.

egories. It includes 20,000 images in the training set and 2,000 images in the validation set. In our experiments, Mamba is employed as the backbone for UperNet [93], with initialization from ImageNet pre-trained weights. The training is conducted using the AdamW optimizer [46, 59] with learning rate of 0.0001 and batch size of 16 for 160,000 iterations. Our implementation utilizes the PyTorch [62] and mmsegmentation [16] libraries. Experiments are performed on four GPUs of NVIDIA 4090, with FP16 precision to enhance the training speed.

Results. The performance trend for semantic segmentation on ADE20K is similar to object detection on COCO. MambaOut can outperform some visual Mamba models but cannot match the results of state-of-the-art Mamba models. For instance, LocalVMamba-T [41] surpasses MambaOut-Tiny by 0.5 mIoU in both single scale (SS) and multi-scale (MS) evaluations, further corroborating our *Hypothesis 2* empirically. Additionally, visual Mamba models continue to exhibit notable performance deficits when compared to the more advanced hybrid models that integrate convolution and attention mechanisms, such as SG-Former [71] and TransNeXt [75]. Visual Mamba needs to further showcase its long-sequence modeling strengths by delivering stronger performance in visual segmentation task.

5. Conclusion and discussion

In this paper, we discuss the Mamba mechanism conceptually and conclude that it is ideally suited for tasks with long-sequence and autoregressive characteristics. We analyze common visual tasks against these criteria and argue that introducing Mamba for ImageNet image classification is unnecessary, as it meets neither characteristic. However, the potential of Mamba for visual detection and segmentation tasks, which align with at least the long-sequence characteristic, merits further exploration. To substantiate our claims empirically, we develop MambaOut models that employ Mamba blocks without their core token mixer, SSM. MambaOut surpasses all visual Mamba models on ImageNet, yet it exhibits a notable performance gap compared to state-of-the-art visual Mamba models, thereby validating our assertions.

Limitation. Due to computational resource limitations, this paper only verifies the Mamba concept for common visual tasks, not including visual tasks in specific domains (*e.g.*medical or remote sensing image analysis) and language tasks. We also do not explore further scaling up MmabaOut in this paper while [89] finds that it performs very well.

Impact statement. The influence of MambaOut on our research community will be highly positive. The strength of our research community lies in its embrace of diverse perspectives, which foster the exchange and collision of ideas. It is noteworthy that papers such as "Attention is All You Need" [82], "Attention is NOT All You Need" [25], "Attention is NOT Explanation" [43], and "Attention is NOT NOT Explanation" [87] have all been accepted to top-tier AI conferences, despite their differing claims. Currently, numerous papers apply Mamba to various vision tasks. Our paper serves as a gentle reminder for researchers to carefully analyze the inherent properties of models when applying them. We also frame our claims as "hypotheses" and limit the scope of our study to three commonly used vision tasks, with the intention of encouraging further hypotheses and discussions in our research community. Additionally, our MambaOut models can serve as a natural baseline for future research on visual Mamba.

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